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Professor Roman Słowiński
Co-ordinating Editor-in-Chief
European Journal of Operational Research
Institute of Computing Science, Poznań University of Technology
60-965 Poznań, Poland

Dear Professor Słowiński,

Please find enclosed our manuscript, **"Reproducible Answer Selection over Large Language Model Ensembles, with an Exact Kemeny Reference and a Ground-Truth Audit,"** for consideration as a Theory and Methodology paper in the *European Journal of Operational Research*.

Sampling several answers from a large language model and choosing one is standard practice, but the usual choosing rules are majority vote, which fails when answers do not repeat, or a language-model judge whose verdicts are not reproducible. We formulate selection as an integer program over a judge-derived pairwise matrix and a self-consistency term, solved exactly, with the exact Kemeny–Young winner as a reference for its 5-approximate aggregate, and we derive a closed-form condition on the judge's pairwise accuracy under which judge-based selection beats plurality voting.

The contribution is an audit rather than a claim of superiority. On six benchmarks (532 problems), paired exact tests show no benefit of judge-based selection with a single generator and a significant harm with heterogeneous generators, which we trace to a length-scale mismatch in the objective and a generator-specific judge preference. The measured pairwise accuracy of the judge sits at chance, far below the derived threshold. The work fits the journal because it applies integer programming, rank aggregation and decision-analytic evaluation to a decision problem of growing practical weight, and it builds on this journal's literature on Kemeny aggregation and ensemble selection. Code, stored runs and tables are archived on Zenodo (DOI 10.5281/zenodo.24112045).

Sincerely,

Aris Dressino